```cpp
#include <iostream>

#include <vector>

#include <chrono>


using namespace std;

using namespace chrono;


// Linear Search

int findLinear(const vector<int>& numbers, int target)

{

  int position = -1;


  for (size_t i = 0; i < numbers.size(); i++)

  {

    if (numbers[i] == target)

    {

      position = i;

      break;

    }

  }


  return position;

}


// Binary Search

int findBinary(const vector<int>& numbers, int target)

{

  int left = 0;

  int right = numbers.size() - 1;


  while (left <= right)
```

```
    {
        int center = (left + right) / 2;

        if (numbers[center] == target)
        {
            return center;
        }
        else if (target > numbers[center])
        {
            left = center + 1;
        }
        else
        {
            right = center - 1;
        }
    }

    return -1;
}

int main()
{
    const int SIZE = 100000;
    vector<int> numbers(SIZE);

    // Store sorted values
    for (int i = 0; i < SIZE; i++)
    {
        numbers[i] = i + 1;
    }
```

```cpp
int target;
cout << "Enter the value to search: ";
cin >> target;

int result;

// Measure Linear Search
auto begin = high_resolution_clock::now();
result = findLinear(numbers, target);
auto finish = high_resolution_clock::now();

cout << "\n----- Linear Search -----" << endl;

if (result >= 0)
   cout << "Value found at index: " << result << endl;
else
   cout << "Value not found." << endl;

cout << "Execution Time: "
    << duration_cast<microseconds>(finish - begin).count()
    << " microseconds" << endl;

// Measure Binary Search
begin = high_resolution_clock::now();
result = findBinary(numbers, target);
finish = high_resolution_clock::now();

cout << "\n----- Binary Search -----" << endl;

if (result >= 0)
   cout << "Value found at index: " << result << endl;
```

```
    else
        cout << "Value not found." << endl;

    cout << "Execution Time: "
         << duration_cast<microseconds>(finish - begin).count()
         << " microseconds" << endl;

    return 0;
}
```

OUTPUT:

```
Enter the value to search: 152

----- Linear Search -----
Value found at index: 151
Execution Time: 2 microseconds

----- Binary Search -----
Value found at index: 151
Execution Time: 1 microseconds



=== Code Execution Successful ===
```